Amir Reza Naderi Yaghouti

PhD Researcher in Medical Image Analysis | Marie Skłodowska-Curie Doctoral Fellow

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PROFILE

Final-stage PhD researcher developing deep-learning and foundation-model methods for glaucoma from fundus images and structural MRI, with a focus on interpretability and rigorous evaluation on small clinical datasets. First author of four peer-reviewed journal articles.

RESEARCH APPOINTMENT

Marie Skłodowska-Curie Doctoral Researcher (DC9), EGRET-AAA | OVGU Magdeburg (DE) & UMC Groningen (NL) | 2024 – present

Supervisors and references: Prof. Michael B. Hoffmann (OVGU), Prof. Nomdo M. Jansonius (UMCG); contact details on request

- **Retinal AI:** CNN, vision-transformer and foundation-model pipelines for glaucoma detection and staging from fundus photographs; interpretability via anatomy-guided region ablation.
- **Structural MRI:** MedSAM-based optic-chiasm segmentation and volumetry showing reduced chiasm volume in advanced glaucoma.

EDUCATION

- **PhD, Visual Neuroscience** (joint), OVGU Magdeburg / UMC Groningen · 2024 – present
- **MSc, Biomedical Engineering**, Islamic Azad University (SRB), Tehran · 2018 – 2021 · GPA 18.43/20; thesis awarded highest grade
- **BSc, Biomedical Engineering**, Islamic Azad University (SRB), Tehran · 2014 – 2018 · Final-year GPA 18.09/20

PUBLICATIONS (FIRST AUTHOR ON ALL)

1. **Naderi Yaghouti, A. R.,** Shalbaf, A., Alizadehsani, R., et al. Artificial intelligence for ovarian cancer detection with medical images: a review of the last decade (2013–2023). *Archives of Computational Methods in Engineering*, 32, 4093–4124 (2025).
2. **Naderi Yaghouti, A. R.,** & Shalbaf, A. Classification of nonalcoholic fatty liver grades using pre-trained convolutional neural networks and a random forest classifier on B-mode ultrasound images. *Journal of Biomedical Physics and Engineering*, 15(5), 437–450 (2025).
3. **Naderi Yaghouti, A. R.,** Zamanian, H., & Shalbaf, A. Machine learning approaches for early detection of non-alcoholic steatohepatitis based on clinical and blood parameters. *Scientific Reports*, 14, 2442 (2024).
4. **Naderi Yaghouti, A. R.,** Shalbaf, A., & Maghsoudi, A. Automatic classification of non-alcoholic fatty liver using texture features from ultrasound images. *Tehran University Medical Journal*, 79(1), 10–17 (2021).

- **Naderi Yaghouti, A. R.,** et al. Deep learning-based segmentation of structural MRI reveals reduced optic chiasm volume in advanced glaucoma. Under review at *NeuroImage: Clinical* (November 2026).
- **Naderi Yaghouti, A. R.,** et al. Discrimination of benign and malignant ovarian tumors from routine laboratory biomarkers using a tabular foundation model. Under review at *Scientific Reports*.

CONFERENCE PRESENTATIONS

CLINICCAI 2026 (MICCAI satellite), Strasbourg · Oral

EVER 2026, Florence · Poster

ECVP 2025, Mainz · Poster

SERVICE, TEACHING AND COMMUNITY

Peer reviewer, Acta Ophthalmologica

EGRET-AAA website team (2026), egret-aaa.eu

Teaching Assistant, Signals and Systems, Islamic Azad University

SKILLS AND LANGUAGES

Methods: segmentation, classification, transfer learning, vision transformers, foundation models, radiomics, explainable AI

Computing: Python, PyTorch, MONAI, TensorFlow/Keras, scikit-learn, MATLAB, Git, LaTeX, GPU/HPC

Languages: English (fluent), German (basic)